Python basics

Question 1: What is Python and why is it widely used in Data Analytics? Mention any three reasons.

Answer:

Python is a high-level, interpreted programming language that makes it easy for anyone — to learn and use. It's known for its simple syntax, readability, and versatility — meaning it can be used for almost anything: from building websites to analyzing data or training AI models.

Python is often described as a "beginner-friendly language", but it's also powerful enough to be used by professionals and large tech companies.

- Beginner Friendly as Python has simple, readable syntax. Also it is a good start for non-programmers.
- Easy Syntax as Python uses natural language structure where indentation makes the code readable.
- Python has Large Community Support.
- Powerful Libraries like Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn

Question 2: Explain the difference between List and Tuple in Python.

Answer:

List

A list is an ordered collection of items, where each item can be of any data type (e.g., integers, strings).

Lists are written using square brackets [].

Key Features:

- Mutable: Items can be added, removed, or changed.
- Indexed: Items can be accessed using their position in the list.

Tuple

- Definition: A tuple is an immutable sequence of items, meaning the elements cannot be changed after the tuple is created.

  Tuples are written using ().

- Key Features:
  - Ordered: Elements maintain their sequence.
  - Immutable: No addition, removal, or modification of items.
  - Allows Duplicates: Like lists, tuples can have duplicate values.

Why Use Tuples?

- Use tuples when you need a fixed collection of items that won't change.
- Suitable for data that must remain constant, such as coordinates, configuration settings, or dictionary keys.

Question 3: What is a function in Python? Why are functions useful?

Answer:

A **function** is a block of reusable code that performs a specific task.

Instead of writing the same code repeatedly, we can create a function once and call it whenever needed.

Advantages

- Code Reusability
- Reduces repetition
- Makes code easier to understand
- Easy to debug and maintain



Question 4: Write a Python program to take a user's name as input and print a greeting message.

Answer:

```
#user name
name = ("Enter your name,:")
print("hello","Roshni")

# print a greeting message
print("welcome in python")
```

```
hello Roshni
welcome in python
```

Question 5: Write a Python program to check whether a number is even or odd using conditional statements.

Answer:

```
number = int(input("enter a number:"))

if number % 2 == 0:
   print("even number")
else:
   print("odd number")
```

```
enter a number:12
even number
```

```
number = int(input("enter a number:"))

if number % 2 == 0:
   print("even number")
else:
   print("odd number")
```

```
enter a number:9
odd number
```

Question 6: Write a program to print numbers from 1 to 10 using a loop.

Answer:

```
for i in range (1,11):
 print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

Question 7: Create a list of five numbers and print the maximum number from the list.

Answer:

```
#MAX
number = (12,45,89,100)

maximum =max(number)

print("maximum number", maximum)
```

```
maximum number 100
```

Question 8: Write a Python program to remove duplicate values from a list using a set.

Answer:

```
numbers =[9,7,8,9,7,8]

unique_numbers = list(set(numbers))

print(unique_numbers)
```

```
[8, 9, 7]
```

Question 9: Write a function that returns the square of a number.

Answer:

```
def sqaure(num):
  return num * num

result = sqaure(9)
print(result)
```

```
81
```

```
def sqaure (num):
  return num*num

result =sqaure(7)
print(result)
```

```
49
```

Question 10: Write a Python program to count how many times a number appears in a list. Example List.

[2,3,4,2,5,2].

Answer:

```
number=[2,3,4,2,5,2]
search = int(input("enter a number,"))
count = number.count(search)
print(search,"appears",count,"times")
```

```
enter a number,2
2 appears 3 times
```

```
number=[2,3,4,2,5,2]
search = int(input("enter a number,"))
count = number.count(search)
print(search,"appears",count,"times")
```

```
enter a number,3
3 appears 1 times
```

```
number=[2,3,4,2,5,2]
search = int(input("enter a number,"))
count = number.count(search)
print(search,"appears",count,"times")
```

```
enter a number,5
5 appears 1 times
```